Water-Efficient Smart Irrigation Scheduling Using Dynamic Programming

CSA0620:DESIGN AND ANALYSIS OF ALGORITHM

Guided By:

Dr Parimala Kanaga Devan K

Presented By:

K. Srinivasa Reddy (192525452)

P. Manoj Kumar Reddy (192511220)

B. Girish Kumar (192210500)

B. Purna Kishor (192472171)

Abstract

- Agriculture requires large amounts of water for crop production.
- Improper irrigation leads to water wastage and increased energy consumption.
- This project develops a smart irrigation scheduling application.
- It uses Dynamic Programming, Greedy and Knapsack algorithms.
- The system optimizes water allocation and estimates pumping energy.

Objectives

- Minimize unnecessary water consumption in agriculture.
- Allocate available water among multiple crop plots efficiently.
- Apply DAA algorithms to optimize irrigation schedules.
- Estimate pumping duration and energy consumption.
- Compare algorithm performance using measurable results.

Problem Statement

- Farmers may distribute water without considering individual crop requirements.
- Limited water resources can lead to insufficient irrigation.
- Unplanned irrigation may increase water and electricity usage.
- Manual scheduling can be inefficient for multiple crop plots.
- An intelligent scheduling system is needed to allocate water effectively.

Literature Review

Author / Year	Research focus	Key findings
Togneri et al. (2023)	Data-driven water requirement estimation	Reviewed data-driven techniques for estimating crop water needs in smart irrigation. ScienceDirect
Bounajra et al. (2024)	Intelligent irrigation management	Identified the importance of crop, soil, climate and field characteristics in irrigation planning. ScienceDirect
A Systematic Review and Meta-Analysis of Intelligent Irrigation Systems (2024)	Sensor-based and AI irrigation	Classified research into field measurements, remote sensing and AI methods. DOI
Jaiswal et al. (2025)	IoT-based smart drip irrigation	Reviewed irrigation automation, optimization and the need for standardized benchmarks. Springer Nature Link
A Review of Smart Irrigation (2025)	Smart irrigation technologies	Discussed optimization methods, automation and challenges in practical adoption.

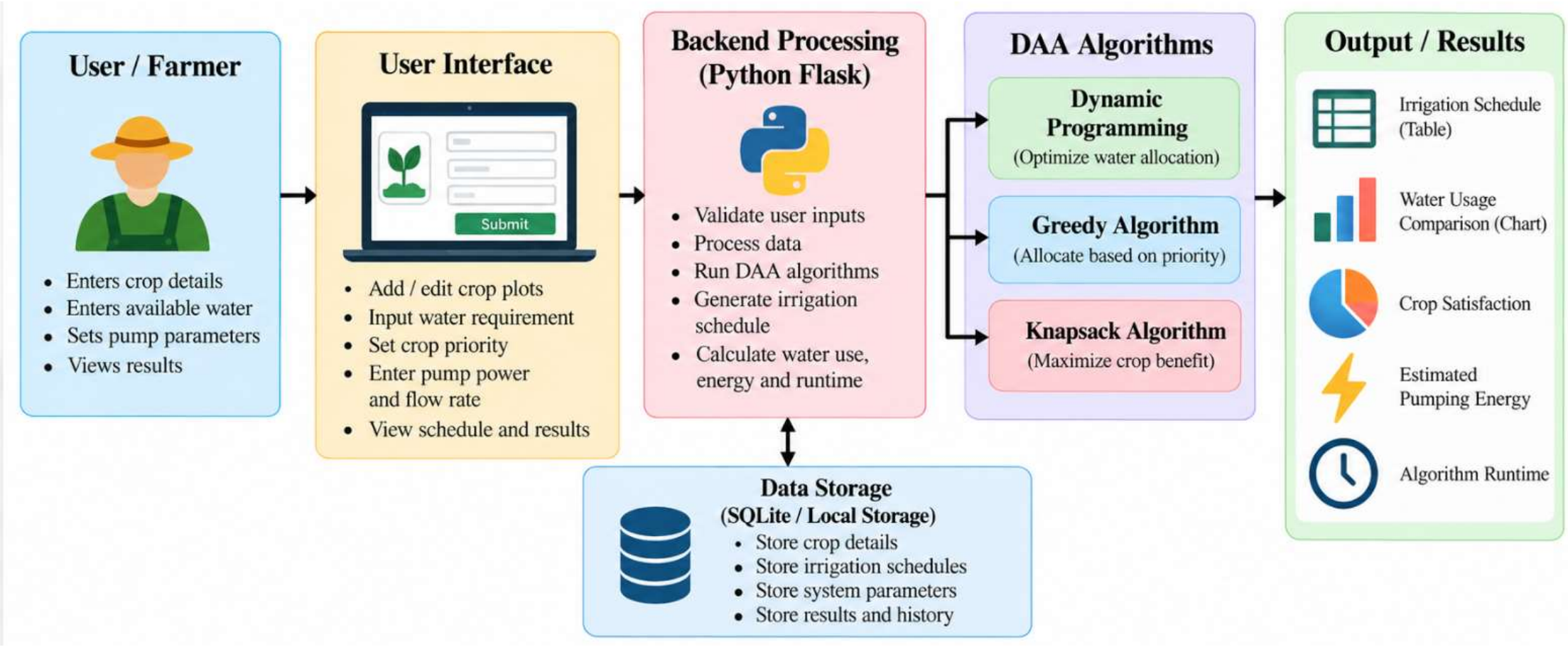
Expected Issues

- Limited Water Availability: Available water may be insufficient to satisfy all crop requirements.
- Input Data Accuracy: Incorrect crop-water requirements can produce inefficient schedules.
- Algorithm Complexity: Dynamic Programming may require more memory and computation for large datasets.
- Energy Estimation: Pumping energy estimates depend on flow rate, pump power and operating conditions.
- Algorithm Comparison: Greedy and Knapsack may produce different results depending on the allocation rules and optimization objectives

Possible Solutions

- Prioritize crops based on urgency and water requirements.
- Validate user inputs before running the algorithms.
- Use suitable water-allocation units to reduce computation.
- Apply consistent formulas for pumping duration and energy.
- Compare all algorithms using the same input data and constraints.

System Architecture



Module 1: Crop and Water Requirement Management

- Enter crop type, plot area and water requirements.
- Add, edit and delete crop plot details.
- Set crop priority and estimated benefit.
- Enter available water and pump parameters.

Module 2: Dynamic Irrigation Scheduling

- Apply Dynamic Programming to optimize water allocation.
- Use the Greedy algorithm for priority-based allocation.
- Apply the Knapsack algorithm to maximize crop benefit.
- Generate an irrigation timetable with water quantity and pump duration.

Module 3: Water and Energy Optimization

- Calculate total water consumption and water savings.
- Estimate pump operating duration for each crop plot.
- Calculate pumping energy using pump power and operating time.
- Identify water shortages and unmet crop requirements.

Module 4: Performance Analysis and Results Dashboard

- Compare the results of all three algorithms.
- Measure algorithm execution time.
- Display crop-water satisfaction and water allocation charts.
- Generate irrigation reports and export the timetable as a CSV file.

Expected Outcomes

- Efficient allocation of limited water resources.
- Reduced avoidable water wastage.
- Better identification of crop-water shortages.
- Estimation of pumping duration and energy consumption.
- Clear comparison of algorithm efficiency and performance.

Conclusion

- The project applies DAA algorithms to an agricultural problem.
- It generates efficient irrigation schedules for multiple crop plots.
- It evaluates water use, crop satisfaction and estimated energy consumption.
- It compares Dynamic Programming, Greedy and Knapsack algorithms.
- The application supports water conservation and sustainable agriculture.

References

- 1. A. B. Gundim et al., “Precision irrigation trends and perspectives: A review,” Ciência Rural, vol. 53, no. 8, 2023.
- 2. H. Gao, L. Zhangzhong, W. Zheng, and G. Chen, “How can agricultural water production be promoted? A review on machine learning for irrigation,” Journal of Cleaner Production, vol. 414, Art. no. 137687, 2023.
- 3. A. Bounajra, K. El Guemmat, K. Mansouri, and F. Akef, “Towards efficient irrigation management at field scale using new technologies: A systematic literature review,” Agricultural Water Management, vol. 295, Art. no. 108758, 2024.
- 4. “A systematic review and meta-analysis of intelligent irrigation systems,” IEEE Access, vol. 12, pp. 128285–128304, 2024.
- 5. N. Jaiswal, T. Vijay Kumar, and C. Shukla, “Smart drip irrigation systems using IoT: A review of architectures, machine learning models, and emerging trends,” Discover Agriculture, vol. 3, Art. no. 253, 2025.

Thank You